

## Alpha decay 2

Let our hypothesis be that the nucleus produces alpha particles at an average rate  $\nu$  which then tunnel with probability  $e^{-2\gamma}$ . Hence the decay rate  $\lambda$  is

$$\lambda = \nu e^{-2\gamma}$$

The half-life  $\tau_{1/2}$  is

$$\tau_{1/2} = \frac{\log 2}{\lambda} = \frac{\log 2}{\nu} e^{2\gamma}$$

We seek to calculate  $\tau_{1/2}$  from  $\gamma$ . The problem is that  $\nu$  is only known empirically so we use the following modified Geiger-Nuttall formula

$$\log \tau_{1/2} = 3.682 \frac{Z}{\sqrt{E}} - 120.8$$

where  $\tau_{1/2}$  is time in seconds and  $E$  is energy in MeV.

For the alpha decay process  ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th}$  we have

$$Z = 90$$

$$E = 4.27 \text{ MeV}$$

Hence

$$\tau_{1/2} = 1.49 \times 10^{17} \text{ s} = 4.72 \times 10^9 \text{ yr}$$

The actual half-life of  ${}^{238}\text{U}$  is

$$4.46 \times 10^9 \text{ yr}$$

The observed linearity of  $\log \tau_{1/2}$  and  $Z/\sqrt{E}$  confirms the tunneling hypothesis for alpha decay.

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